TEAM 03

AI POWERED DATA ANALYSIS REMOTE INTERNSHIP

Excelerate

TEAM MEMBERS

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OVERVIEW

This project analyzes learner participation and engagement across opportunities such as courses, internships, events, and competitions.

The project workflow includes:

- Data Cleaning and Preprocessing
- Feature Engineering
- Exploratory Data Analysis (EDA)
- Churn Analysis and Predictive Modeling

DATASET DESCRIPTION

The dataset contains:

- Learner demographics
- Opportunity details
- Application timelines
- Status information

Total records: **8,558**
Total variables: **16**

Examples of columns:

- Learner SignUp DateTime
- Apply Date
- Gender
- Country
- Opportunity Start Date
- Status Code

WEEK 1: DATA CLEANING OBJECTIVES

The purpose of data cleaning was to:

- Fix invalid timestamps
- Convert date columns to datetime
- Handle missing values
- Standardize categorical values
- Remove duplicates
- Validate logical errors

DATA CLEANING STEPS

Major cleaning steps included:

- Fixing broken timestamps (e.g., “24:00”)
- Converting object columns to datetime
- Standardizing gender and country values
- Handling missing values
- Removing duplicate records
- Validating age and date logic

Dataset completeness after cleaning: 91.9%

FEATURE ENGINEERING

New features were created to improve analysis.

Examples:

- Age_Current
- Age_Group
- Days_to_Apply
- Application_Speed
- Opportunity_Duration_Days
- Engagement_Score
- Engagement_Level
- Region

These features helped measure learner behavior and engagement.

ENGAGEMENT SCORE

Engagement Score was calculated using:

- Application speed
- Age alignment
- Signup time

Formula:

Engagement Score = Average of Speed_Score, Age_Score, and Time_Score

Engagement levels:

- Highly Engaged
- Moderately Engaged
- Average
- Low

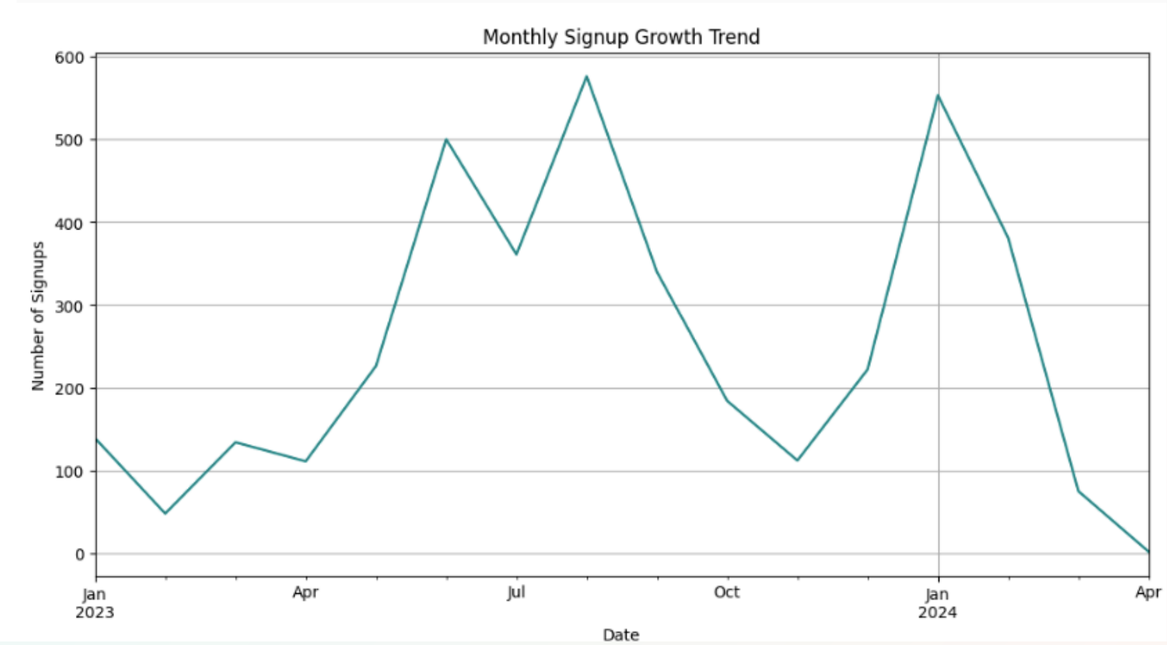
WEEK 2: EXPLORATORY DATA ANALYSIS

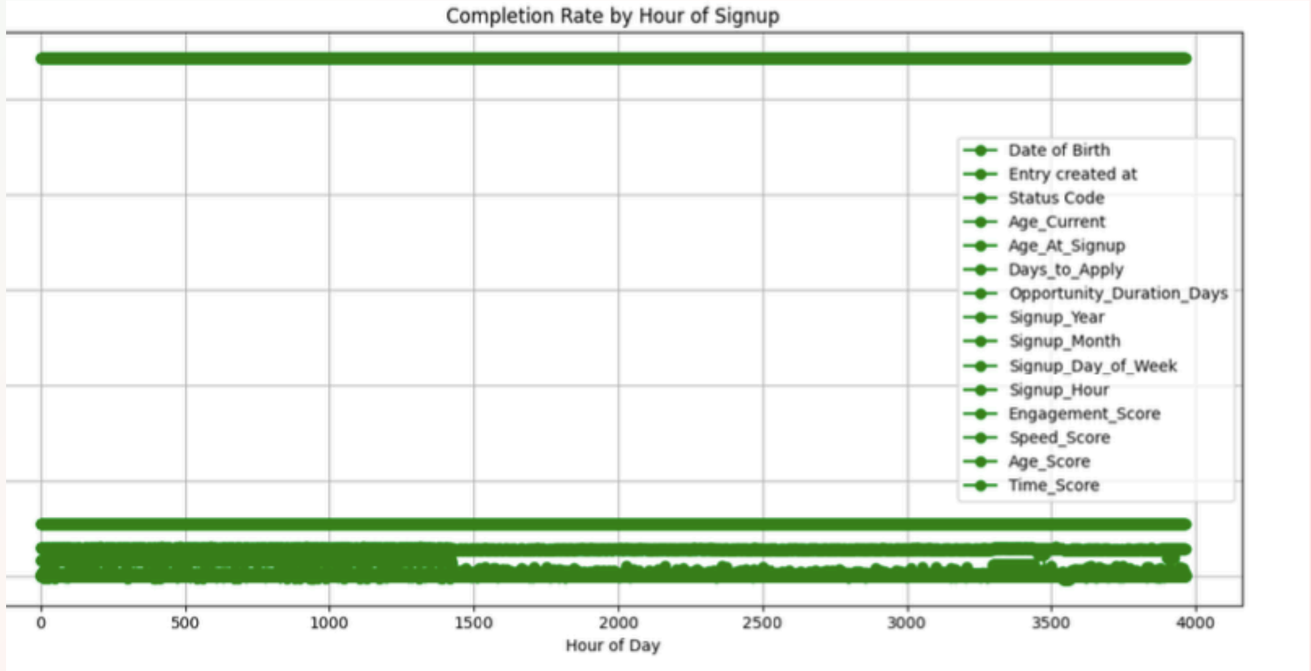
The goal of EDA was to identify:

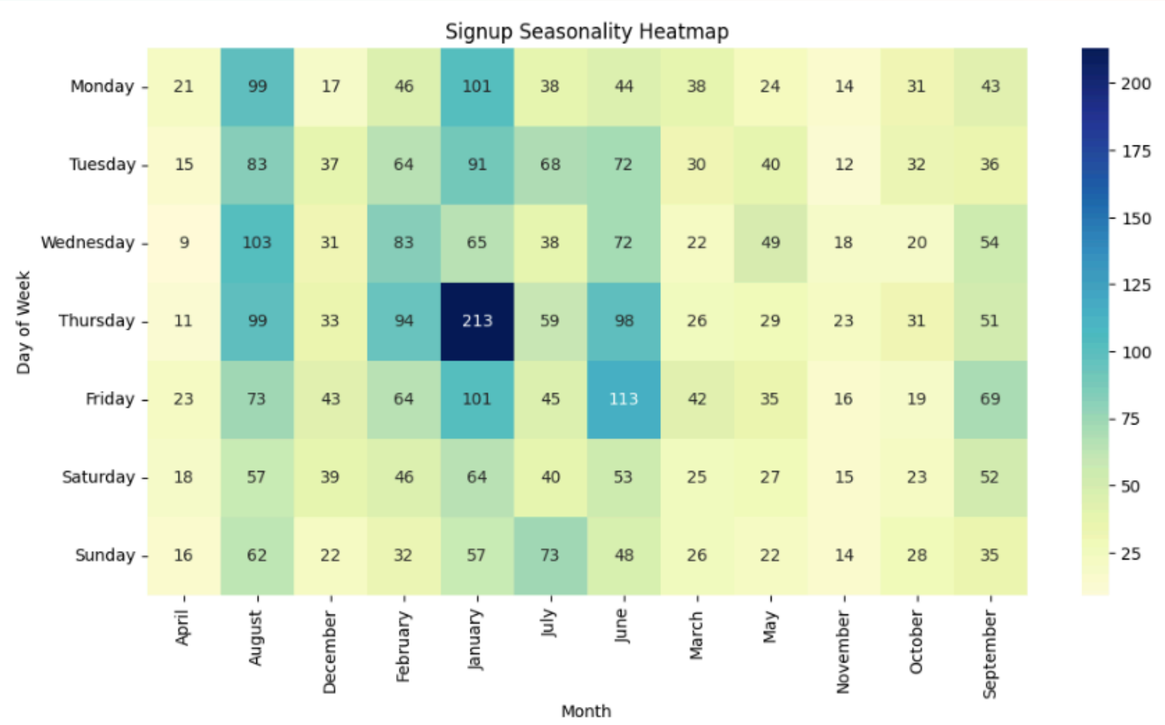
- Signup trends
- Seasonal patterns
- Completion behavior
- Engagement relationships
- Outliers and anomalies

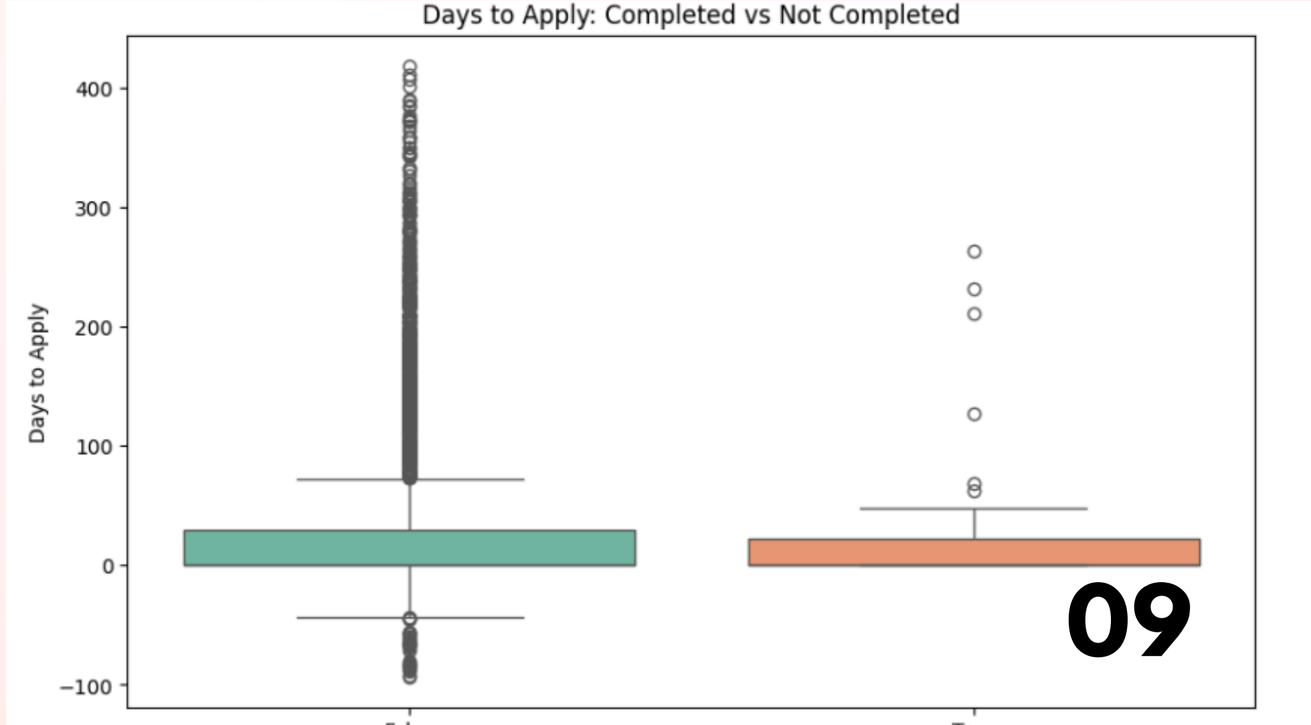
Visualizations were created using:

- Line charts
- Heatmaps
- Boxplots
- Scatter plots

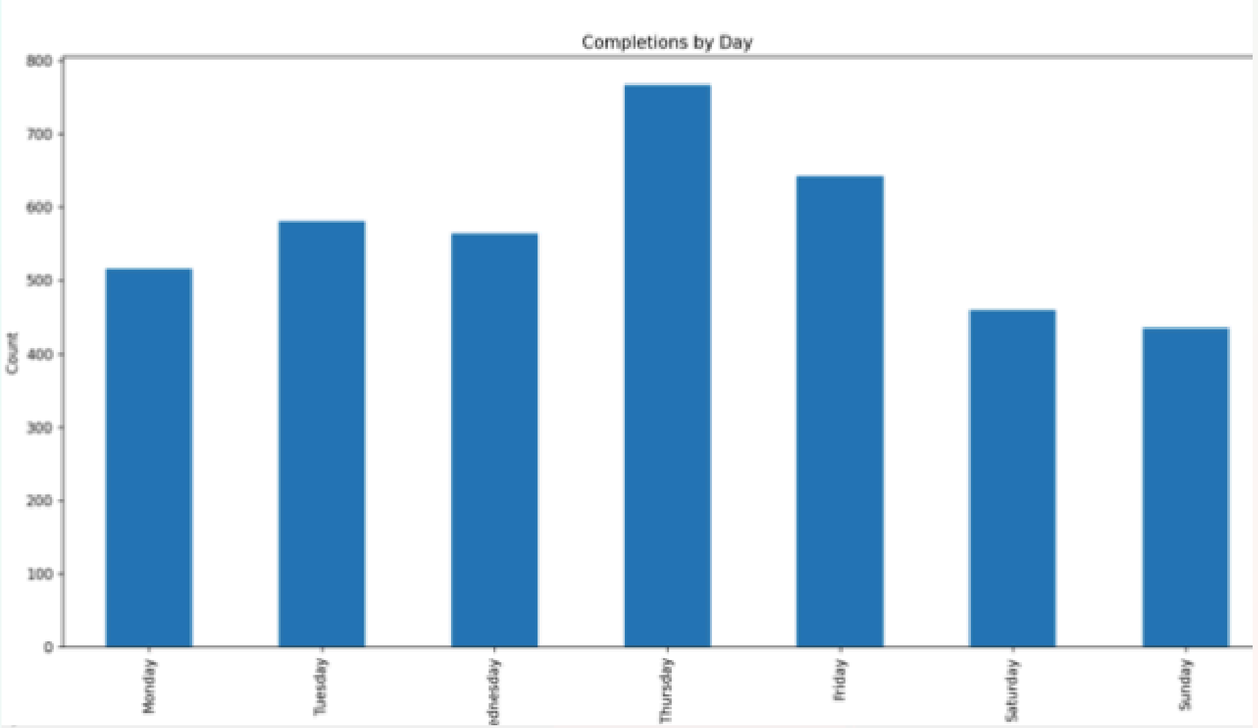

Monthly Signup Growth Trend
Number of Signups
600
500
400
300
200
100
0
Jan 2023
Apr
Jul
Oct
Jan 2024
Apr
Date

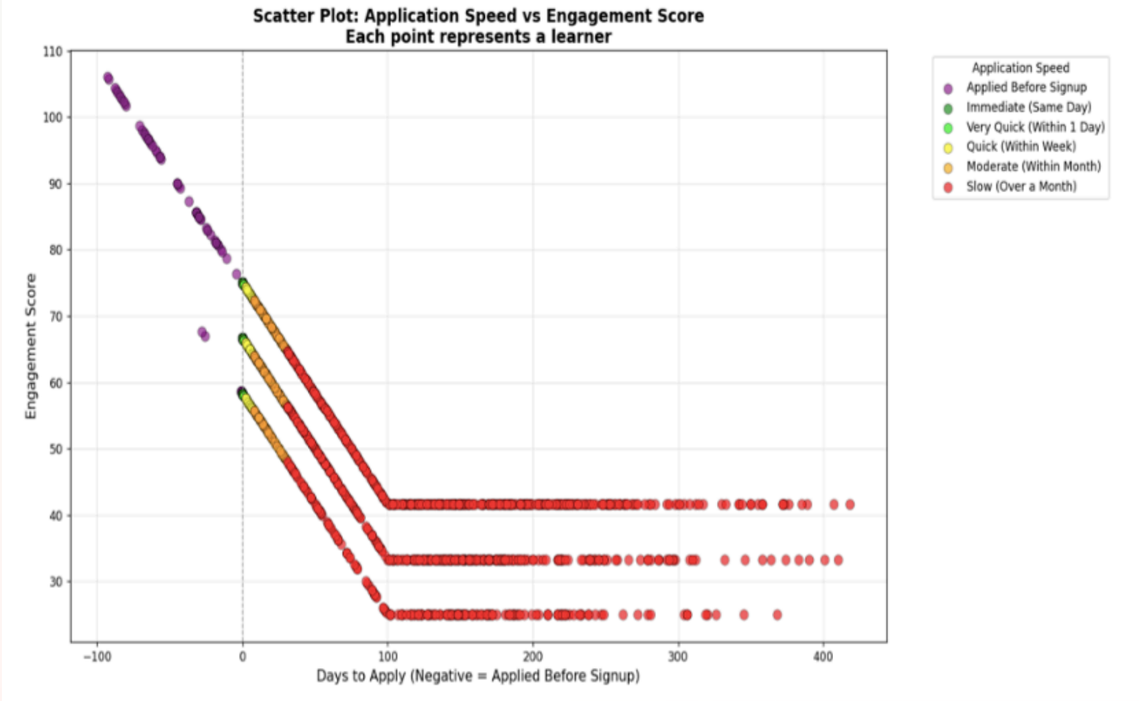

Completion Rate by Hour of Signup
Date of Birth
Entry created at
Status Code
Age_Current
Age_At_Signup
Days_to_Apply
Opportunity_Duration_Days
Signup_Year
Signup_Month
Signup_Day_of_Week
Signup_Hour
Engagement_Score
Speed_Score
Age_Score
Time_Score
0
500
1000
1500
2000
2500
3000
3500
4000
Hour of Day

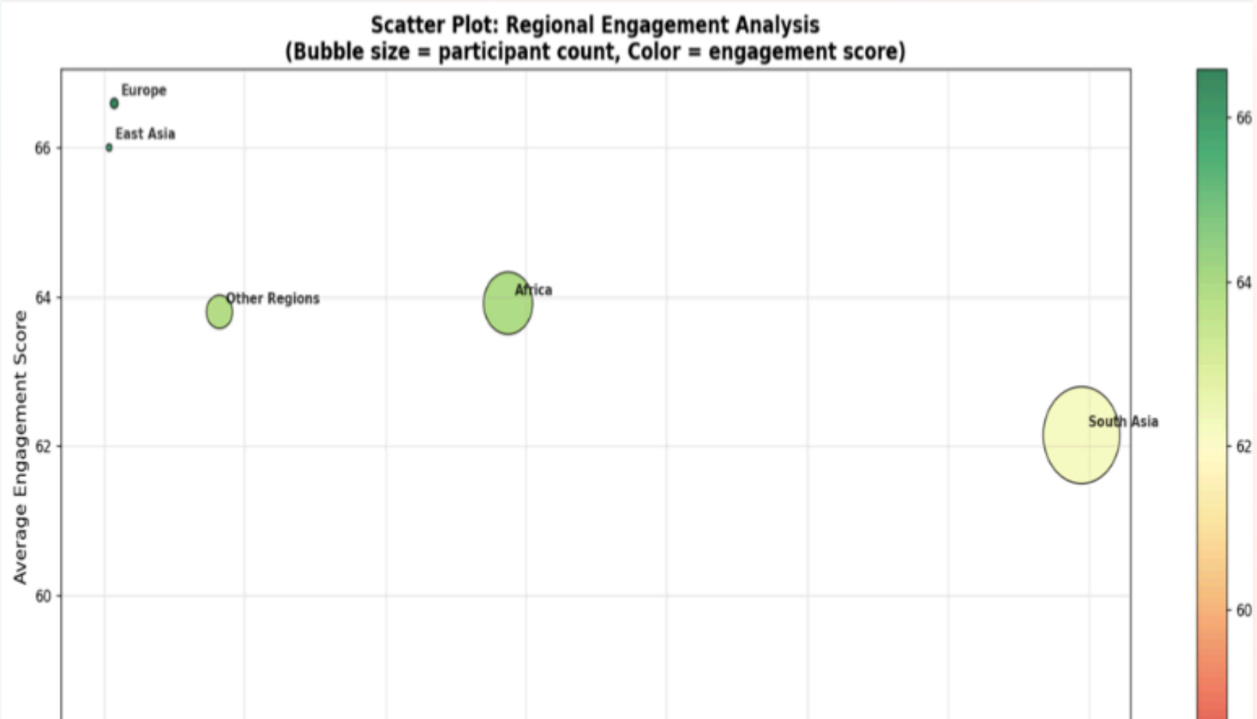

Signup Seasonality Heatmap
Day of Week
Monday 21 99 17 46 101 38 44 38 24 14 31 43
Tuesday 15 83 37 64 91 68 72 30 40 12 32 36
Wednesday 9 103 31 83 65 38 72 22 49 18 20 54
Thursday 11 99 33 94 213 59 98 26 29 23 31 51
Friday 23 73 43 64 101 45 113 42 35 16 19 69
Saturday 18 57 39 46 64 40 53 25 27 15 23 52
Sunday 16 62 22 32 57 73 48 26 22 14 28 35
April
August
December
February
January
July
June
March
May
November
October
September
Month
200
175
150
125
100
75
50
25

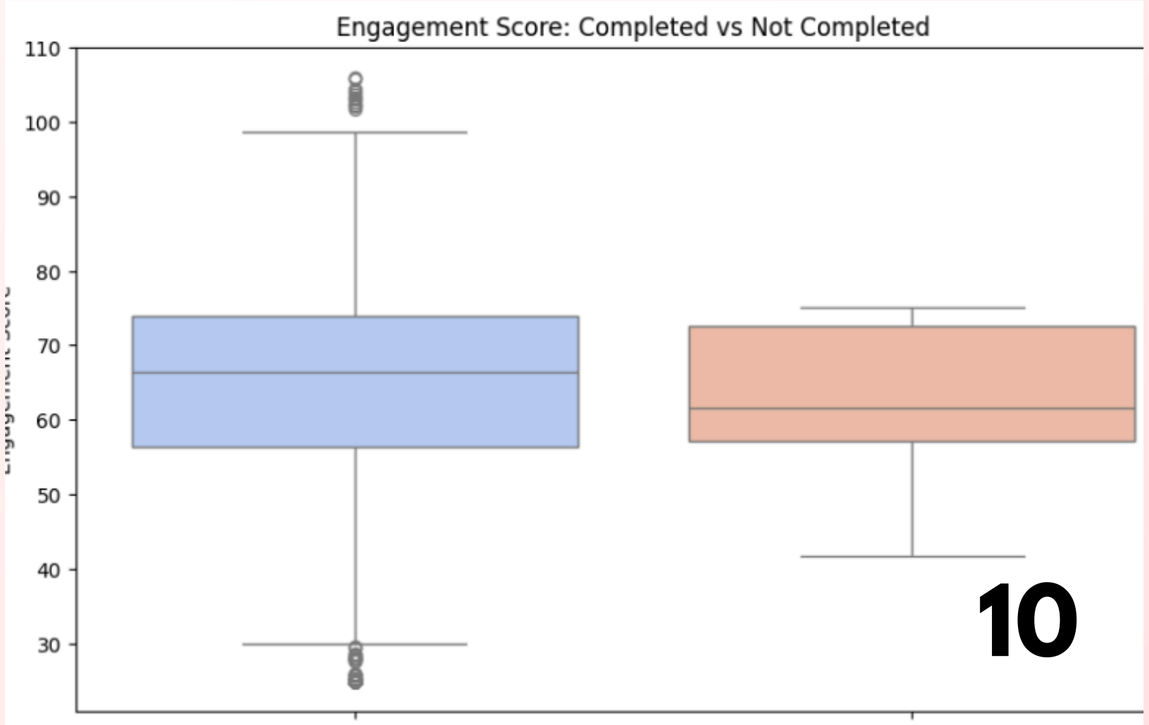

Days to Apply: Completed vs Not Completed
Days to Apply
400
300
200
100
0
−100

09


Completions by Day
Count
800
700
600
500
400
300
200
100
0
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday


Scatter Plot: Application Speed vs Engagement Score
Each point represents a learner
Engagement Score
Days to Apply (Negative = Applied Before Signup)
Application Speed
Applied Before Signup
Immediate (Same Day)
Very Quick (Within 1 Day)
Quick (Within Week)
Moderate (Within Month)
Slow (Over a Month)


Scatter Plot: Regional Engagement Analysis
(Bubble size = participant count, Color = engagement score)
Average Engagement Score
Europe
East Asia
Other Regions
Africa
South Asia


Engagement Score: Completed vs Not Completed

OBSERVATIONS

Signup Trends:

- Overall growth in signups observed
- Clear seasonal fluctuations
- Noticeable campaign-driven spikes

Completion Behavior

- Higher engagement leads to higher completion rates
- Faster applicants are more likely to complete
- Completion varies by signup hour

Seasonality Patterns

- Peak signup months identified
- Weekdays are more active than weekends
- Consistent low-activity periods

Outlier Analysis

- Small group with extreme delays
- Maximum application delay: 620 days
- Outliers represent 2.2% of records

Timing and engagement are key drivers of completion

WEEK 3: CHURN ANALYSIS

Churn refers to learners who:

- Delay applying
- Show low engagement
- Do not complete applications

Goal:

Identify early warning signals of learner drop-off.

Churn Indicators

- Low engagement score
- Long application delays
- Withdrawal status
- Certain signup times

PREDICTIVE & ANALYTICAL MODELING

Application Behavior Model:

- Analyzes how quickly learners apply
- Identifies timing and seasonality patterns
- Highlights early vs delayed application behavior

Cohort Planning & Forecast Model

- Forecasts future application demand
- Uses historical signup trends with confidence intervals
- Supports proactive cohort and resource planning

Engagement Scoring Model

- Segments learners into High, Medium, and Low engagement
- Driven by application speed and opportunity category
- Excludes age due to incomplete demographic data

Models are designed for interpretability and decision support.

KEY INSIGHTS & RECOMMENDATIONS

Key Findings

- Application activity shows clear seasonal and timing patterns
- Early applicants consistently demonstrate higher engagement
- Engagement level strongly influences completion likelihood
- Completion rates vary by signup hour and weekday
- A small group of users shows extreme application delays, indicating drop-off risk

Recommended Actions

- Target peak signup hours for outreach and announcements
- Trigger reminders for applicants with delayed application timelines
- Provide targeted support for low-engagement learners
- Adopt region-specific engagement strategies rather than uniform messaging
- Monitor signup trends in real time to anticipate cohort demand

Focusing on timing, speed, and engagement signals enables proactive intervention and more effective cohort planning.

DASHBOARD 1

Learner Engagement & Application Insights Dashboard

High-level summary of learner applications and engagement

3K	4K	2K	38.0%	61.04	33.89
Total Learners	Total Applications	Accepted Applications	Acceptance Rate (%)	Avg Engagement Score	Avg Days to Apply

Country
- Afghani...
- Algeria
- America...
- Australia
- Azerbaij...
- Banglad...
- Belarus
- Bhutan

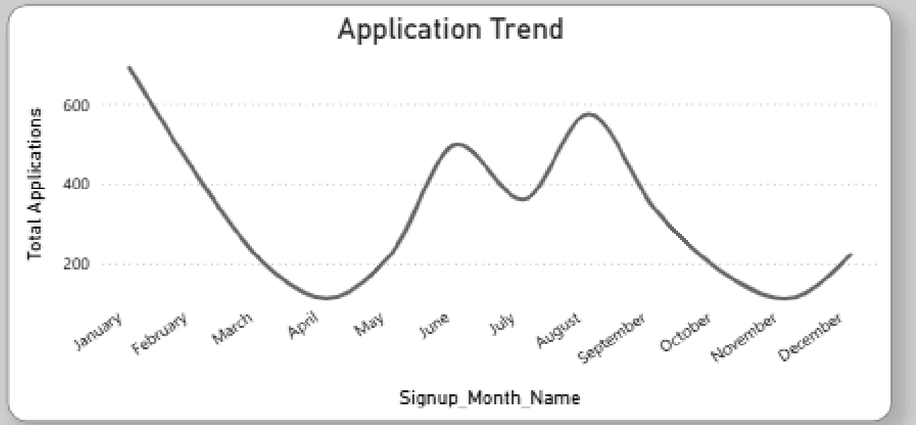
Signup Season
- Fall
- Spring
- Summer
- Winter

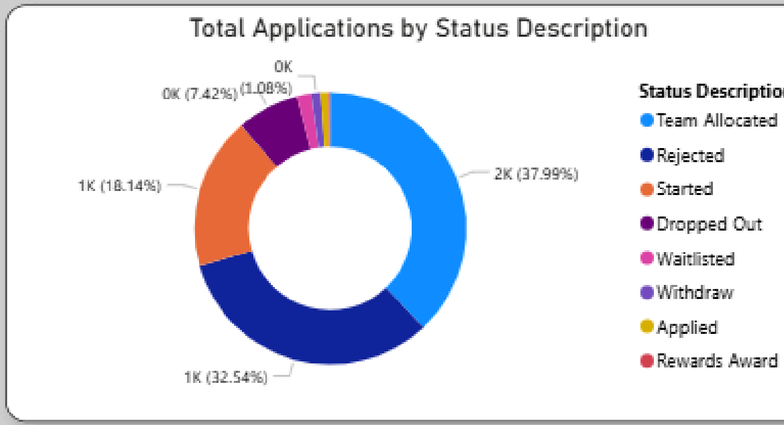
Time_of_Day
- Afternoon
- Evening
- Morning
- Night

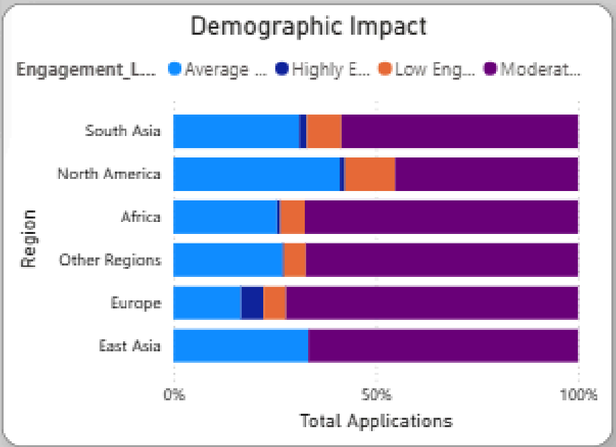
Region
- Africa
- East Asia
- Europe
- North Amer...
- Other Regio...
- South Asia

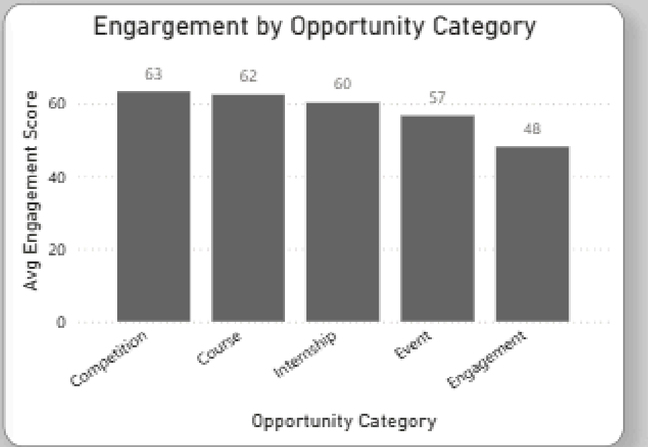
Opportunity Category
- Competition
- Course
- Engagement
- Event
- Internship

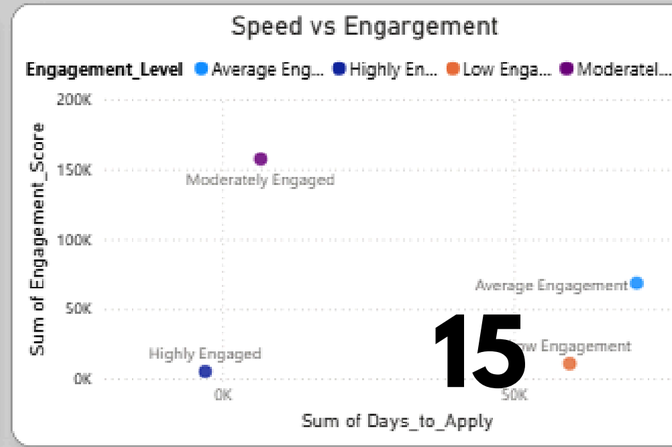
Engagement Level
- Average En...
- Highly Enga...
- Low Engag...
- Moderately ...











DASHBOARD 2

Application Behavior Model

Understanding learner application timing and speed patterns

33.89
Avg Days to Apply

2K
Immediate Applicants

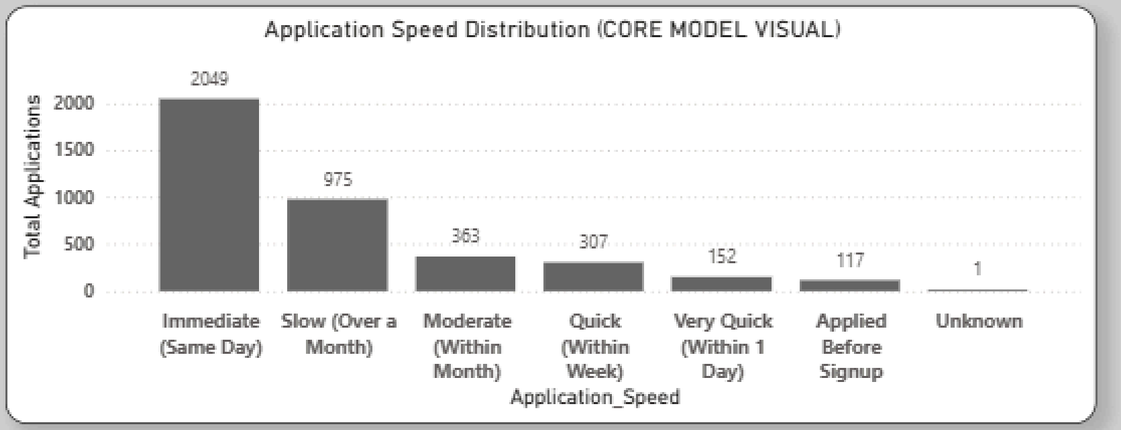
363
Moderate Applicants

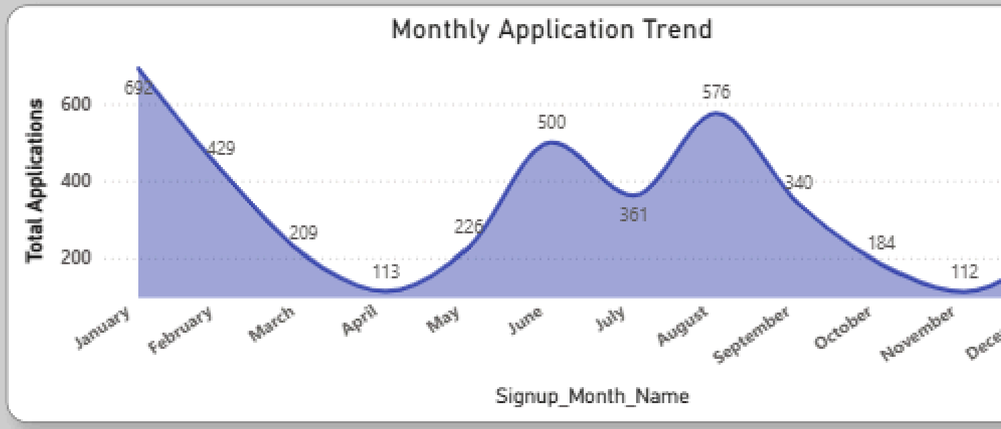
975
Slow Applicants

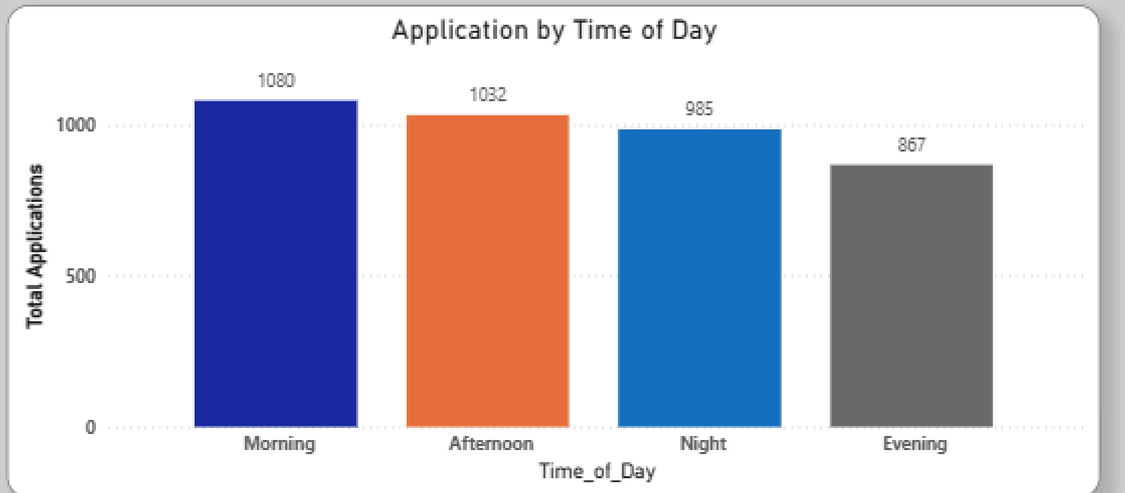
Region
- Africa
- East Asia
- Europe
- North America
- Other Regions
- South Asia

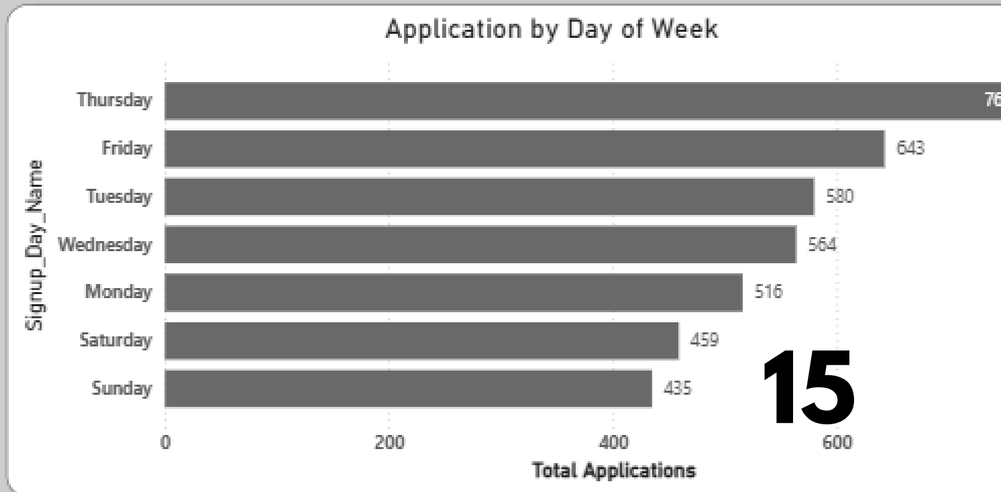
Signup_Season
- Fall
- Spring
- Summer
- Winter

Opportunity Category
- Competition
- Course
- Engagement
- Event
- Internship









DASHBOARD 3

Engagement Scoring Model

Segmenting learners based on engagement intensity and key drivers

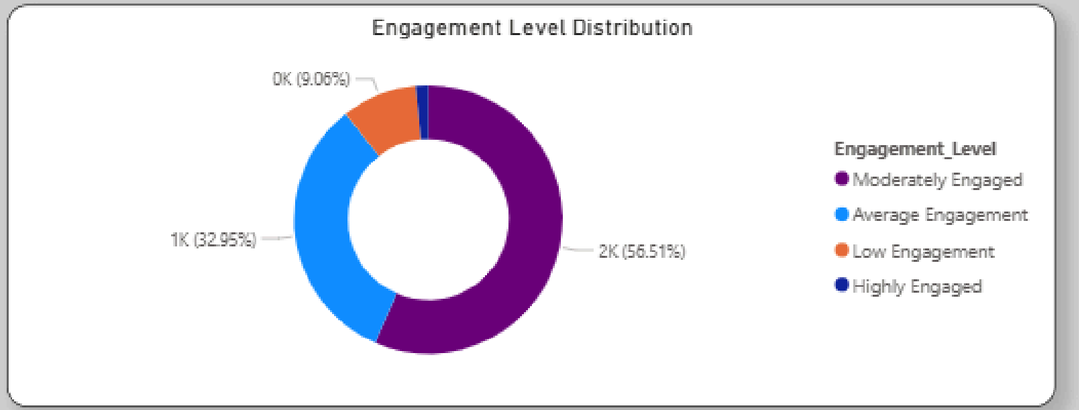
61.04
Avg Engagement Score

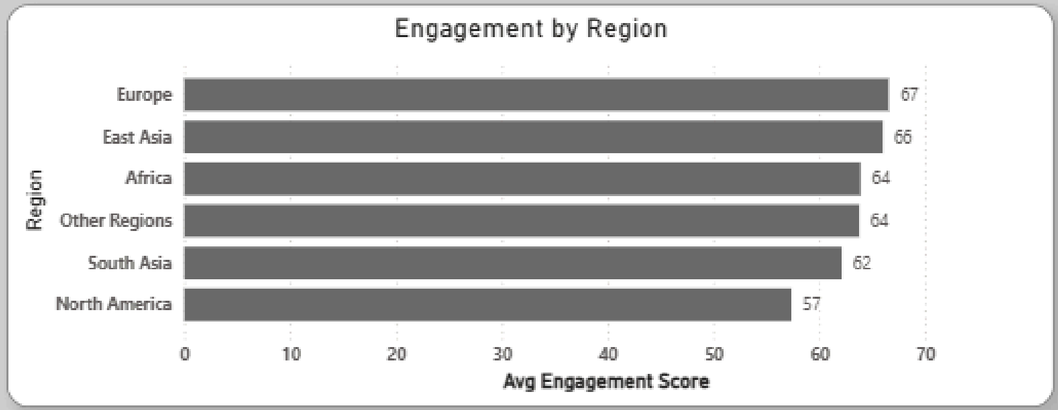
2K
Medium Engagement Learners

59
High Engagement Learners

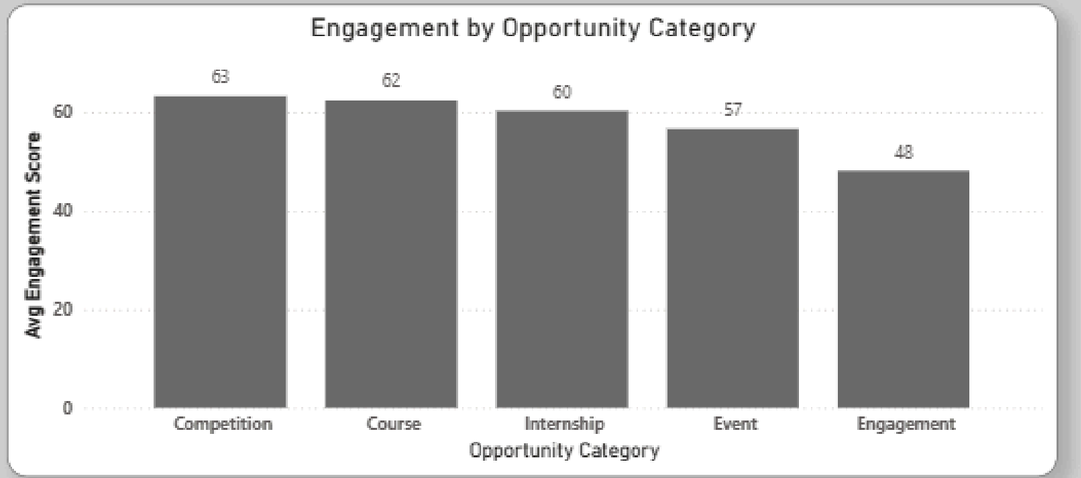
359
Low Engagement Learners

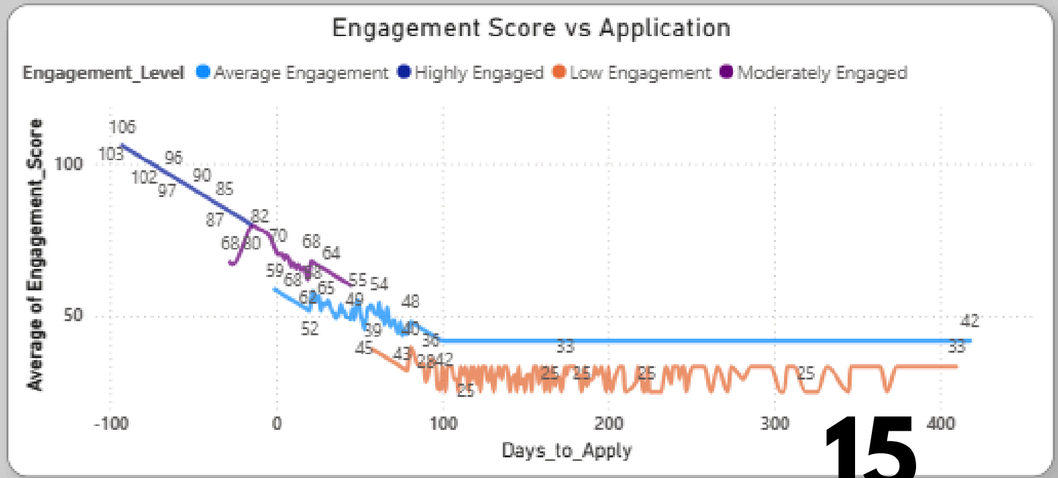
Region
- Africa
- East Asia
- Europe
- North America
- Other Regions
- South Asia





Opportunity Category
- Competition
- Course
- Engagement
- Event
- Internship





Engagement levels derived from composite engagement scoring model.

DASHBOARD 4

Cohort Planning & Forecast Model

Using historical trends to support future cohort planning

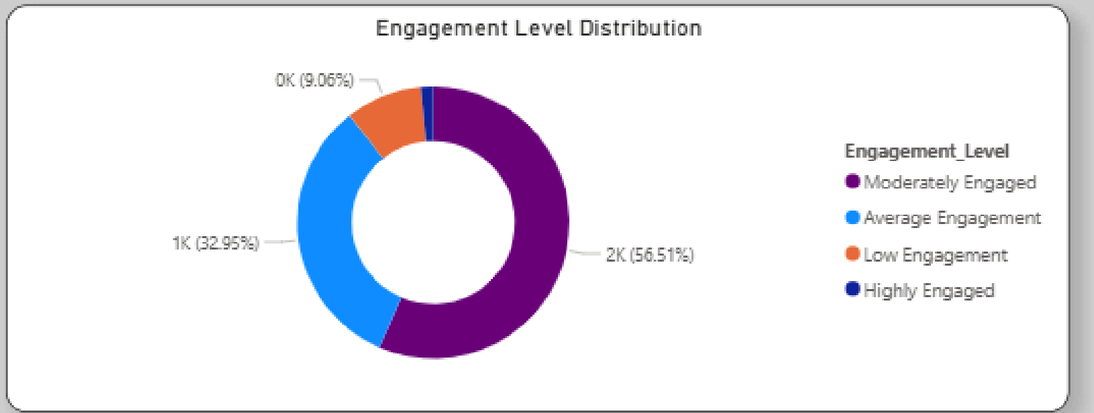
61.04
Avg Engagement Score

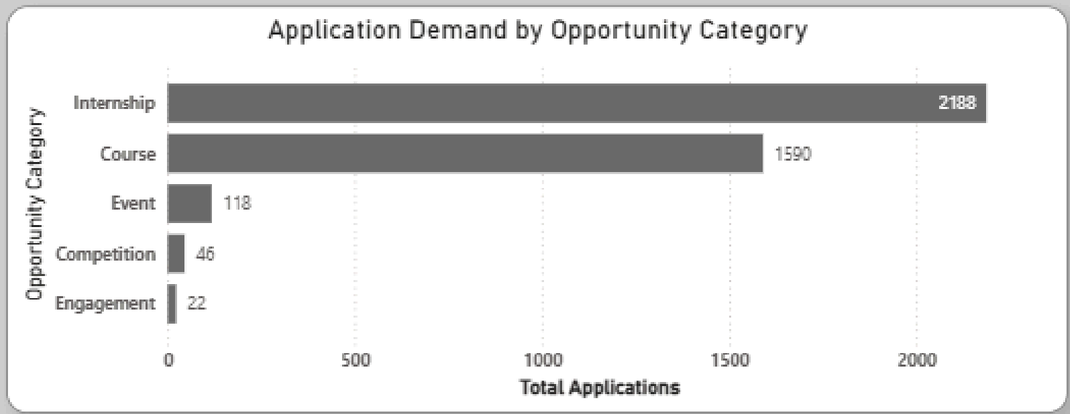
4K
Total Applications

59
High Engagement Learners

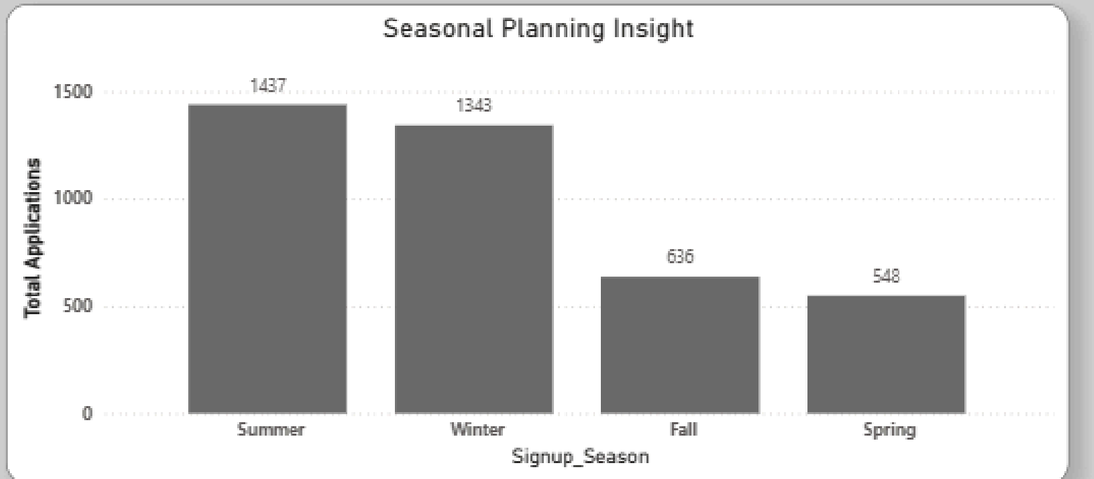
359
Low Engagement Learners

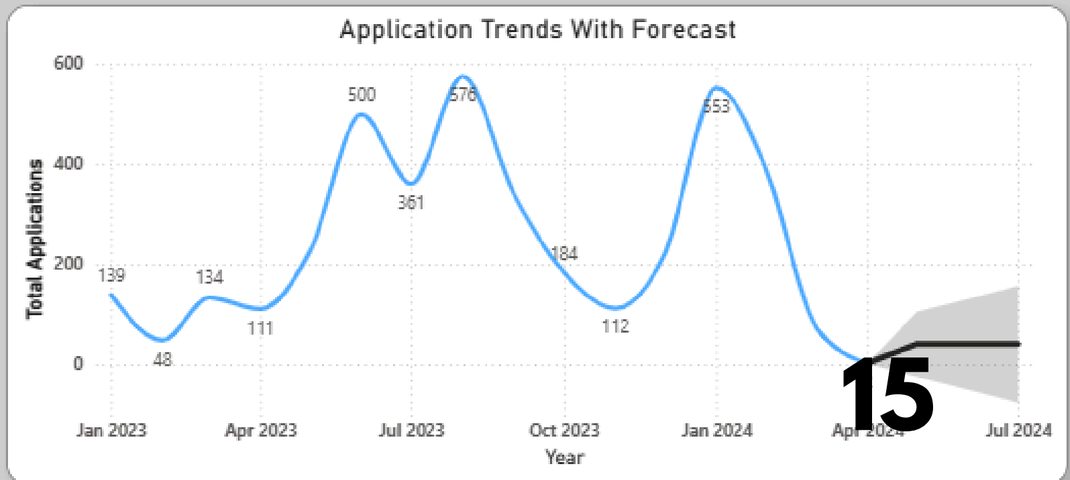
Region
- Africa
- East Asia
- Europe
- North America
- Other Regions
- South Asia





Opportunity Category
- Competition
- Course
- Engagement
- Event
- Internship





THANK YOU